EB-JEPA for EEG — What We Did and What It Shows

A precise but compact summary of our original contributions

EB-JEPA EEG track

June 2026

Setting. All experiments use TUAB v3.0.1 (2 717 train / 276 eval recordings, 19 channels, 200 Hz, 10-second windows, patient-disjoint split). The encoder is pretrained with SSL (no labels), then a frozen logistic regression probe is trained on the labelled set. Primary metric: per-recording BACC (mean-pool 16 windows per patient).

1 What the framework provides (not our contribution)

The **EB-JEPA library** provides the two-view SSL training loop, the VICReg and SIGReg loss functions, the Projector MLP, the RNNPredictor, and the data loading pipeline.

The **baseline Conv1D encoder** (4 blocks of strided Conv1d, BatchNorm, GELU, 0.4M parameters) produces a per-window embedding via global average pooling of the final feature map.

Standard augmentations (Gaussian noise σ=0.1, amplitude jitter ±20%, channel drop p=0.2) are the default pipeline.

The baseline with these defaults and VICReg scores **0.796 BACC per-recording**.

2 Contribution 1 — Corruption augmentation (+2.3 pp, our main lever)

What we added. On top of the standard augmentations, each view receives:

- 20 % of time steps **masked to zero** (contiguous or random)
- 20 % of time steps replaced by $\pm 6\sigma$ **spike outliers**

This is inspired by the observation that real EEG artefacts (electrode pop, muscle noise, electrode drift) produce exactly these patterns. The encoder is forced to learn what persists through corruption — the clinically meaningful signal structure — rather than low-level amplitude statistics.

What it shows.

Augmentation	BACC_{REC} (3-seed)	AUROC
Standard only	0.796	0.888
+ **Corruption**	**0.819 ± .004**	**0.900 ± .006**

The gain is $+\mathbf{0.023}$, reproducible across 3 seeds (± 0.004). No other modification — more data, wider encoder, longer training, different objective — exceeds this. Corruption is the single most impactful design decision in our work.

3 Contribution 2 — Novel finding: EEG prefers weak VICReg invariance

What VICReg invariance means. The VICReg loss has three terms:

$$\mathcal{L} = \lambda_{\text{inv}}\, \|z_1 - z_2\|^2 + \lambda_{\text{var}}\, V(z_1, z_2) + \lambda_{\text{cov}}\, C(z_1, z_2)$$

The invariance term $\|z_1 - z_2\|^2$ pushes the two views together. The variance term V prevents collapse by keeping each output dimension spread out. The covariance term C decorrelates dimensions. The original VICReg paper (Bardes et al., ICLR 2022, vision SSL) uses $\lambda_{\text{inv}} = 25$. The EB-JEPA library default is $\lambda_{\text{inv}} = 1$.

What we found. Running the same setup (corruption, Conv1D) with both settings, 3 seeds each:

λ_{inv}	BACC_{REC}	AUROC
1 (default)	**0.819 ± .004**	**0.900 ± .006**
5	0.820	0.893
10	0.814	0.900
25 (paper-correct)	0.789 ± .005	0.882 ± .004

Endpoints 3-seed; intermediates single-seed.

The sweet spot is $\lambda \in \{1, 5\}$ (0.819–0.820, essentially tied), with a clear drop at λ=10 and a large drop at λ=25 (**3 pp,** $\approx 6\sigma$). The ordering is not strictly monotone at the top but the overall trend is unambiguous: more invariance $\Rightarrow$ lower BACC for EEG.

Why this matters. With $\lambda_{\text{inv}} = 1$, the anti-collapse terms (variance + covariance) dominate the invariance term. The encoder is pushed to *spread representations* more than it is pushed to make two views identical. For short-window clinical EEG with a small conv encoder, this regime produces more discriminative features.

This is not a bug fix — it is a finding. To our knowledge, no prior work has systematically ablated VICReg's invariance coefficient for EEG SSL.

4 Contribution 3 — NaiveJEPA: collapse-free prediction without regularisation

What we built. NaiveJEPA splits a single EEG window in time at ratio 0.5:

- Context ($t = 0 \to T/2$): online encoder → mean pool → MLP predictor
- Target ($t = T/2 \to T$): EMA copy of encoder → mean pool (no gradient)
- Loss: $\text{MSE}(\hat{z}_{\text{tgt}},\ z_{\text{tgt}})$

No anti-collapse term is needed. Because context and target see disjoint time spans, collapsing to a constant would drive the MSE loss to the variance of the target distribution (not zero), so collapse is architecturally prevented.

What it shows.

Objective	BACC_{REC}	Hyperparameters
VICReg + corruption	0.819 ± .004	3 loss weights + EMA + augmentation policy
NaiveJEPA	**0.779**	**EMA only (1 hyperparameter)**

The gap to VICReg is only 0.040 pp with a fraction of the design complexity. NaiveJEPA also achieves **0.389 BACC on TUEV** (6-class transfer), beating ST-Transformer (0.3385) and CNN-Transformer (0.4100) trained end-to-end on TUEV — without ever seeing a TUEV label.

5 Contribution 4 — Cross-task transfer: TUEV without TUEV labels

What we did. After SSL pretraining on TUAB (binary, no labels used), we froze the encoder and trained a logistic regression probe on TUEV (6-class event detection): spike-and-slow-wave, generalised periodic discharge, lateralised periodic discharge, eye movement, artefact, background.

What it shows.

Method	TUEV BACC	Trained on TUEV?
ST-Transformer (supervised)	0.3385	Yes, end-to-end
CNN-Transformer (supervised)	0.4100	Yes, end-to-end
VICReg (TUAB SSL + probe)	**0.382**	Probe only
NaiveJEPA (TUAB SSL + probe)	**0.389**	Probe only
FFCL (supervised)	0.4551	Yes, end-to-end
BIOT pretrained (supervised)	0.5009	Yes, end-to-end

Our SSL encoder beats two fully-supervised models on the target task *while never having trained on the target task at all.* This is the strongest evidence that the representations are general: they encode structure of EEG that is relevant beyond the binary abnormality task.

6 Failure modes we documented

EEGPT architecture conflict

EEGPT's cross-attention pools 19 channels into 1 token per time step, leaving only 10 temporal tokens. Channel-drop augmentation randomly zeroes out channels in view v_1 but not v_2, making the two views structurally incompatible *before the transformer sees them.* The encoder cannot learn invariance to a transformation that changes the token count. Result: representations below random initialisation. Fix: remove channel-drop → TUEV BACC 0.182 → 0.358.

Evaluation protocol mismatch

Per-recording BACC is ≈ 5 pp higher than per-window for the same encoder. Most published literature (BIOT, LaBraM, FEMBA) reports per-window. We initially compared our per-recording 0.836 to LaBraM's per-window 0.814 and concluded we matched LaBraM — this is invalid. Our best per-window score is 0.775 (SIGReg + corruption), 3.9 pp below LaBraM-Base.

Spectral loss confound

An auxiliary DDSP-style multi-scale spectral loss appeared to give +0.040 BACC when added to VICReg ($0.796 \rightarrow 0.836$). A sweep of the spectral coefficient $\{0.05, 0.10, 0.20, 0.30\}$ shows the gain disappears: all coefficients give 0.795–0.813, within single-seed noise of 0.819 (no spectral). The original 0.836 was confounded: single seed, and the coefficient was chosen by looking at the evaluation set.

7 What the binding constraint is

Every Conv1D variant plateaus at per-recording BACC ≈ 0.82, per-window ≈ 0.77, regardless of:

- Encoder depth or width (scaling)
- 4× more pretraining data (multi-corpus)
- Longer pretraining (not tested beyond 20 epochs, but no convergence gap)
- SSL objective (VICReg, SIGReg, NaiveJEPA all plateau near the same level)

The encoder itself is the ceiling. The 3.9 pp gap to LaBraM-Base is not a loss function problem or a data volume problem — LaBraM uses a learned VQ-codebook patch tokeniser pretrained on 2 500 hours of EEG across 12 institutions. That tokeniser quality, not any SSL objective, is what drives the gap.